

A pencil-and-paper encryption system based on the Enigma

Step 1:

BSCISSAA
IDBYROWS

Wheel setting 1: B
I → O

ABCDEFGHIJKLM N O PQRSTUVWXYZ ABCD ← Hint for the UKW setting

W	A	KRYPTOS	ABCDEF	G	HIJLMN	QUVW	XZ	KRYP	S	W
H	B	RYPTOS	ABCDEFG	H	IJLMN	QUVW	XZK	RYPT	C	H
E	C	YPTOS	ABCDEFGH	I	JLMN	QUVW	XZKR	YPTO	H	E
E	D	PTOS	ABCDEFGHI	J	LMN	QUVW	XZKRY	PTOS	E	E
L	E	TOS	ABCDEFGHIJ	L	MN	QUVW	XZKRYP	TOSA	I	L
	F	OS	ABCDEFGHIJL	M	NQ	UVW	XZKRYPT	OSAB	D	
S	G	S	ABCDEFGHIJLM	N	Q	UVW	XZKRYPTO	SABC	T	S
E	H	A	BCDEFGHIJLMN	Q	U	VW	XZKRYPTOS	ABCD	A	E
T	I	B	CDEFGHIJLMNQ	U	V	W	XZKRYPTOSA	BCDE	B	T
T	J	C	DEFGHIJLMNQU	V	W	X	ZKRYPTOSAB	CDEF	F	T
I	K	D	FGHIJLMNQUV	W	X	Z	KRYPTOSABC	DEFG	G	I
N	L	E	FGHIJLMNQUVW	X	Z	K	RYPTOSABCD	EFGH	J	N
G	M	F	GHIJLMNQUVWX	Z	K	R	YPTOSABCDE	FGHI	K	G
	N	GHIJLMN	QUVWXZ	K	RYPTOS	ABCDEF	GHIJ	L		
1	O	HIJLMN	QUVWXZK	R	YPTOS	ABCDEFG	HIJL	M	2	
	P	IJLMN	QUVWXZKR	Y	PTOS	ABCDEFGH	IJLM	N		
	Q	JLMN	QUVWXZKRY	P	TOS	ABCDEFGHI	JLMN	O		
	R	LMN	QUVWXZKRYP	T	OS	ABCDEFGHIJ	LMNQ	P		
	S	MN	QUVWXZKRYPT	O	S	ABCDEFGHIJL	MNQU	Q		
	T	N	QUVWXZKRYPTO	S	A	BCDEFGHIJLM	NQUV	R		
	U	Q	UVWXZKRYPTOS	A	B	CDEFGHIJLMN	QUVW	U		
	V	U	VWXZKRYPTOSA	B	C	DEFGHIJLMNQ	UVWX	V		
	W	V	WXZKRYPTOSAB	C	D	EFGHIJLMNQU	VWXZ	W		
	X	W	XZKRYPTOSABC	D	E	FGHIJLMNQUV	WXZK	X		
	Y	X	ZKRYPTOSABCD	E	F	GHIJLMNQUVW	XZKR	Y		
	Z	Z	KRYPTOSABCDE	F	G	HIJLMNQUVWX	ZKRY	Z		

UKW EFGHIJKLMNOPQ R STUVWXYZABCD ABCD

Step 2:

UKW setting: R

O → U

ABCDEFGHIJKLM N OPQRSTUVWXYZ ABCD ← Hint for the UKW setting

W	A	KRYPTOS	ABCDEF	G	HIJLMN	QUVWXZ	KRYP	S	W
H	B	RYPTOS	ABCDEFG	H	IJLMN	QUVWXZK	RYPT	C	H
E	C	YPTOS	ABCDEFGH	I	JLMN	QUVWXZKR	YPTO	H	E
E	D	PTOS	ABCDEFGHI	J	LMN	QUVWXZKRY	PTOS	E	E
L	E	TOS	ABCDEFGHIJ	L	MN	QUVWXZKRYP	TOSA	I	L
	F	OS	ABCDEFGHIJL	M	NQ	UVWXZKRYPT	OSAB	D	
S	G	S	ABCDEFGHIJLM	N	Q	UVWXZKRYPTO	SABC	T	S
E	H	A	ABCDEFGHIJLMN	Q	U	VWXZKRYPTOS	ABCD	A	E
T	I	B	CDEFGHIJLMNQ	U	V	WXZKRYPTOSA	BCDE	B	T
T	J	C	DEFGHIJLMNQU	V	W	XZKRYPTOSAB	CDEF	F	T
I	K	D	FGHIJLMNQUV	W	X	ZKRYPTOSABC	DEFG	G	I
N	L	E	FGHIJLMNQUVW	X	Z	KRYPTOSABCD	EFGH	J	N
G	M	F	GHIJLMNQUVWX	Z	K	RYPTOSABCDE	FGHI	K	G
	N	G	HIJLMNQUVWXZ	K	R	YPTOSABCDEF	GHIJ	L	
1	O	H	IJLMNQUVWXZK	R	Y	PTOSABCDEFG	HIJL	M	2
	P	I	JLMNQUVWXZKR	Y	P	TOSABCDEFGH	IJLM	N	
	Q	J	LMNQUVWXZKRY	P	T	OSABCDEFGHI	JLMN	O	
	R	L	MNQUVWXZKRYP	T	O	SABCDEFGHIJ	LMNQ	P	
	S	M	NQUVWXZKRYPT	O	S	ABCDEFGHIJL	MNQU	Q	
	T	N	QUVWXZKRYPTO	S	A	B	CDEFGHIJLM	NQUV	R
	U	Q	UVWXZKRYPTOS	A	B	C	DEFGHIJLMN	QUVW	U
	V	U	VWXZKRYPTOSA	B	C	D	EFGHIJLMNQ	UVWX	V
	W	V	WXZKRYPTOSAB	C	D	E	FGHIJLMNQU	VWXZ	W
	X	W	XZKRYPTOSABC	D	E	F	GHIJLMNQUV	WXZK	X
	Y	X	ZKRYPTOSABCD	E	F	G	HIJLMNQUVW	XZKR	Y
	Z	Z	KRYPTOSABCDE	F	G	H	IJLMNQUVWX	ZKRY	Z

UKW EFGHIJKLMNO P Q R STUVWXYZ ABCD

Step 3:

IMPSESQPAL

OFIALUSION

UHENCODUEA encode U->A

Wheel setting 2: L

U → A

ABCDEFGHIJKLM N OPQRSTUWXYZ ABCD ← Hint for the UKW setting

W	A	KRYPTOS	ABCDEF	G	HIJLMN	QUVW	XZ	KRYP	S	W
H	B	RYPTOS	ABCDEFG	H	IJLMN	QUVW	XZK	RYPT	C	H
E	C	YPTOS	ABCDEFGH	I	JLMN	QUVW	XZKR	YPTO	H	E
E	D	PTOS	ABCDEFGHI	J	LMN	QUVW	XZKRY	PTOS	E	E
L	E	TOS	ABCDEFGHIJ	L	MN	QUVW	XZKRYP	TOSA	I	L
	F	OS	ABCDEFGHIJL	M	NQ	UVW	XZKRYPT	OSAB	D	
S	G	S	ABCDEFGHIJLM	N	Q	UVW	XZKRYPTO	SABC	T	S
E	H	A	ABCDEFGHIJLMN	Q	U	VW	XZKRYPTOS	ABCD	A	E
T	I	B	ABCDEFGHIJLMNQ	U	V	W	XZKRYPTOSA	BCDE	B	T
T	J	C	ABCDEFGHIJLMNQU	V	W	X	ZKRYPTOSAB	CDEF	F	T
I	K	D	ABCDEFGHIJLMNQUV	W	X	Z	KRYPTOSABC	DEFG	G	I
N	L	E	FGHIJLMNQUVW	X	Z	K	RYPTOSABCD	EFGH	J	N
G	M	F	GHIJLMNQUVWX	Z	K	R	YPTOSABCDE	FGHI	K	G
	N	G	HIJLMNQUVWXZ	K	R	YPTOS	ABCDEF	GHIJ	L	
1	O	H	IJLMNQUVWXZK	R	Y	P	TOSABCDEFG	HIJL	M	2
	P	I	JLMNQUVWXZKR	Y	P	T	OSABCDEFGH	IJLM	N	
	Q	J	LMNQUVWXZKRY	P	T	O	SABCDEFGHI	JLMN	O	
	R	L	MNQUVWXZKRYP	T	O	S	ABCDEFGHIJ	LMNQ	P	
	S	M	NQUVWXZKRYPT	O	S	A	BCEFGHIJL	MNQU	Q	
	T	N	QUVWXZKRYPTO	S	A	B	CDEFGHIJLM	NQUV	R	
	U	Q	UVWXZKRYPTOS	A	B	C	DEFGHIJLMN	QUVW	U	
	V	U	VWXZKRYPTOSA	B	C	D	EFGHIJLMNQ	UVWX	V	
	W	V	WXZKRYPTOSAB	C	D	E	FGHIJLMNQU	VWXZ	W	
	X	W	XZKRYPTOSABC	D	E	F	GHIJLMNQUV	WXZK	X	
	Y	X	ZKRYPTOSABCD	E	F	G	HIJLMNQUVW	XZKR	Y	
	Z	Z	KRYPTOSABCDE	F	G	H	IJLMNQUVWX	ZKRY	Z	

UKW EFGHIJKLMNO PQRSTUWXYZ ABCD

Decryption from THEWA(X)Y to BE(R)LIN

(without UKW)

WATJKLUD Cipher
JKLMNOPQ Settings 2
TUVWXYZA Settings 1
OIFXBZPM Interm.
11111111 Masking 0,-1,-2
THEWAXYL Plain

(UKW setting N)

IAWINFBNYPVTT Cipher
UVWXYZABCDEFGH Settings 2 ← X on layer two
BCDEFGHIJKLMN Settings 1
SJYFDSSWBFMLN Interm.
2211011001120 Masking 0,-1,-2
THREDOOWBELIN Plain